AI-Powered Fake News Detection Using Text Classification

Author: Kartikeya

Abstract

The rapid growth of digital communication platforms and online news portals has significantly increased the spread of fake news. False information published on social media and news websites can influence public opinion, create panic, manipulate elections, and negatively impact society. Manual verification of every news article is impractical because thousands of articles are published every day. Consequently, automated fake news detection systems based on Natural Language Processing (NLP) and Machine Learning (ML) have become an active area of research.

This project presents an AI-powered fake news detection system capable of classifying news articles as either **Real** or **Fake** using text classification techniques. The system combines multiple stages including data preprocessing, exploratory data analysis, feature extraction, machine learning model training, performance evaluation, and real-time prediction. News articles collected from the Fake.csv and True.csv datasets are cleaned through tokenization, stop-word removal, punctuation removal, number removal, and lemmatization before feature extraction is performed using Term Frequency-Inverse Document Frequency (TF-IDF) and Bag-of-Words techniques.

Four supervised machine learning algorithms—K-Nearest Neighbors (KNN), Logistic Regression, Random Forest, and Multi-Layer Perceptron Neural Network—were trained and evaluated using Accuracy, Precision, Recall, F1-score, Confusion Matrix, Receiver Operating Characteristic (ROC) Curve, Precision-Recall Curve, Cross Validation, and Feature Importance Analysis. Experimental results indicate that the Random Forest classifier achieved the highest overall performance with an accuracy of approximately **99.66%**, outperforming the remaining algorithms. The trained model was further integrated with a prediction module that accepts custom news articles and classifies them into real or fake categories.

The proposed system demonstrates that combining effective preprocessing with appropriate feature engineering and ensemble learning techniques can achieve highly accurate fake news detection while maintaining computational efficiency.

Keywords— Fake News Detection, Machine Learning, Natural Language Processing, Text Classification, Random Forest, Logistic Regression, TF-IDF, Bag of Words.

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1. Introduction

The internet has transformed the way information is created, distributed, and consumed. Millions of people now rely on online news portals, websites, blogs, and social media platforms such as Facebook, X (formerly Twitter), Instagram, and YouTube to obtain information about current events. Although digital communication enables rapid dissemination of information, it also allows false or misleading information to spread quickly without proper verification. Such misinformation, commonly referred to as **fake news**, poses serious challenges to individuals, organizations, and governments worldwide.

Fake news consists of intentionally fabricated or misleading information designed to deceive readers for political, financial, ideological, or social purposes. Unlike genuine journalism, fake news often contains exaggerated headlines, misleading statements, manipulated facts, and emotionally charged language intended to attract public attention. Because such information spreads rapidly through social media sharing and recommendation algorithms, false stories frequently reach a larger audience than verified reports before corrections can be issued.

The increasing influence of fake news has produced several negative consequences. During elections, misinformation campaigns can influence voting behaviour and distort democratic processes. During public health emergencies, false medical information may discourage people from following scientific advice, leading to serious health risks. Financial markets may also experience instability due to misleading reports regarding companies, government policies, or economic conditions. Consequently, detecting fake news has become an important research problem within artificial intelligence and data science.

Traditional approaches to fake news verification primarily rely on professional journalists and independent fact-checking organizations. Although human verification generally produces accurate results, it is time-consuming, expensive, and unable to cope with the enormous volume of information generated every day. Thousands of news articles are published continuously across multiple platforms, making manual verification insufficient for real-time monitoring.

Recent advances in Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) have provided effective techniques for automatically detecting fake news. Machine learning algorithms learn patterns from previously labelled examples and use those learned patterns to classify unseen news articles. Rather than relying solely on manually defined rules, these algorithms analyse writing style, vocabulary, sentence structure, and statistical relationships between words to distinguish genuine news from fabricated information.

Natural Language Processing plays a critical role in this process by transforming raw textual information into structured numerical representations that machine learning algorithms can understand. Before training the classification models, raw news articles require extensive preprocessing to remove unnecessary information such as punctuation, stop words, numbers, URLs, and HTML tags. Techniques such as tokenization and lemmatization normalize textual data while preserving meaningful semantic information. Feature extraction methods including Bag-of-Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF) convert cleaned text into numerical vectors representing word importance within documents.

Several machine learning algorithms have demonstrated promising performance in text classification tasks. Logistic Regression provides a simple yet highly effective linear classifier capable of handling high-dimensional TF-IDF vectors efficiently. K-Nearest Neighbors performs classification based on similarity between documents but often suffers from increased computational complexity as dataset size increases. Random Forest combines multiple decision trees to improve prediction accuracy while reducing overfitting through ensemble learning. Neural Networks learn more complex nonlinear relationships within textual data and often achieve competitive performance when sufficient training data are available.

In this project, an AI-powered fake news detection system has been developed using supervised machine learning techniques. The proposed system follows a complete machine learning pipeline beginning with data loading, preprocessing, exploratory data analysis, feature extraction, model training, evaluation, and deployment of the best-performing model for predicting custom news articles. Multiple performance evaluation metrics including Accuracy, Precision, Recall, F1-score, Confusion Matrix, ROC Curve, Precision-Recall Curve, Cross Validation, and Feature Importance Analysis are employed to compare different machine learning algorithms objectively.

The dataset used in this project consists of two publicly available collections containing fake and real news articles. After preprocessing and cleaning, the dataset is transformed into TF-IDF feature vectors, and four supervised machine learning models are trained using identical training and testing partitions to ensure fair comparison. Experimental results demonstrate that the Random Forest classifier achieved the highest prediction accuracy among all evaluated models while maintaining excellent precision and recall values.

The primary objectives of this project are:

- To develop an automated fake news detection system using machine learning techniques.
- To preprocess textual news data using Natural Language Processing techniques.

- To compare the performance of multiple classification algorithms including KNN, Logistic Regression, Random Forest, and Neural Networks.
- To evaluate classifier performance using multiple statistical metrics rather than relying solely on accuracy.
- To deploy the best-performing model for predicting whether unseen news articles are genuine or fake.

The proposed system provides an efficient, scalable, and highly accurate approach for detecting fake news and demonstrates the practical application of machine learning techniques in addressing one of today's most significant information reliability challenges.

Part 2

2. Dataset Description

The performance of any machine learning model depends significantly on the quality and diversity of the dataset used during training. Since fake news detection is a supervised classification problem, the model requires a labelled dataset containing both genuine and fabricated news articles. In this project, a publicly available Fake News Dataset consisting of **Fake.csv** and **True.csv** files was used to develop and evaluate the classification models. The dataset contains political and world news collected from multiple online news sources and has been widely used in fake news detection research.

The **Fake.csv** file contains fabricated news articles collected from unreliable or satirical sources, whereas **True.csv** contains verified news articles obtained primarily from Reuters and other reliable news agencies. To prepare the dataset for supervised learning, each fake article was assigned a label of **0**, while every genuine article was assigned a label of **1**. Both datasets were then merged into a single dataframe before preprocessing.

After combining the datasets, duplicate news articles were removed to avoid bias during model training. The dataset was shuffled randomly using a fixed random seed so that fake and genuine news articles were uniformly distributed before splitting into training and testing sets. This randomization prevents sequential bias and improves model generalization.

The combined dataset contained **38,646** news articles after duplicate removal. Each record initially consisted of five attributes:

- Title
- News Text

- Subject
- Date
- Label

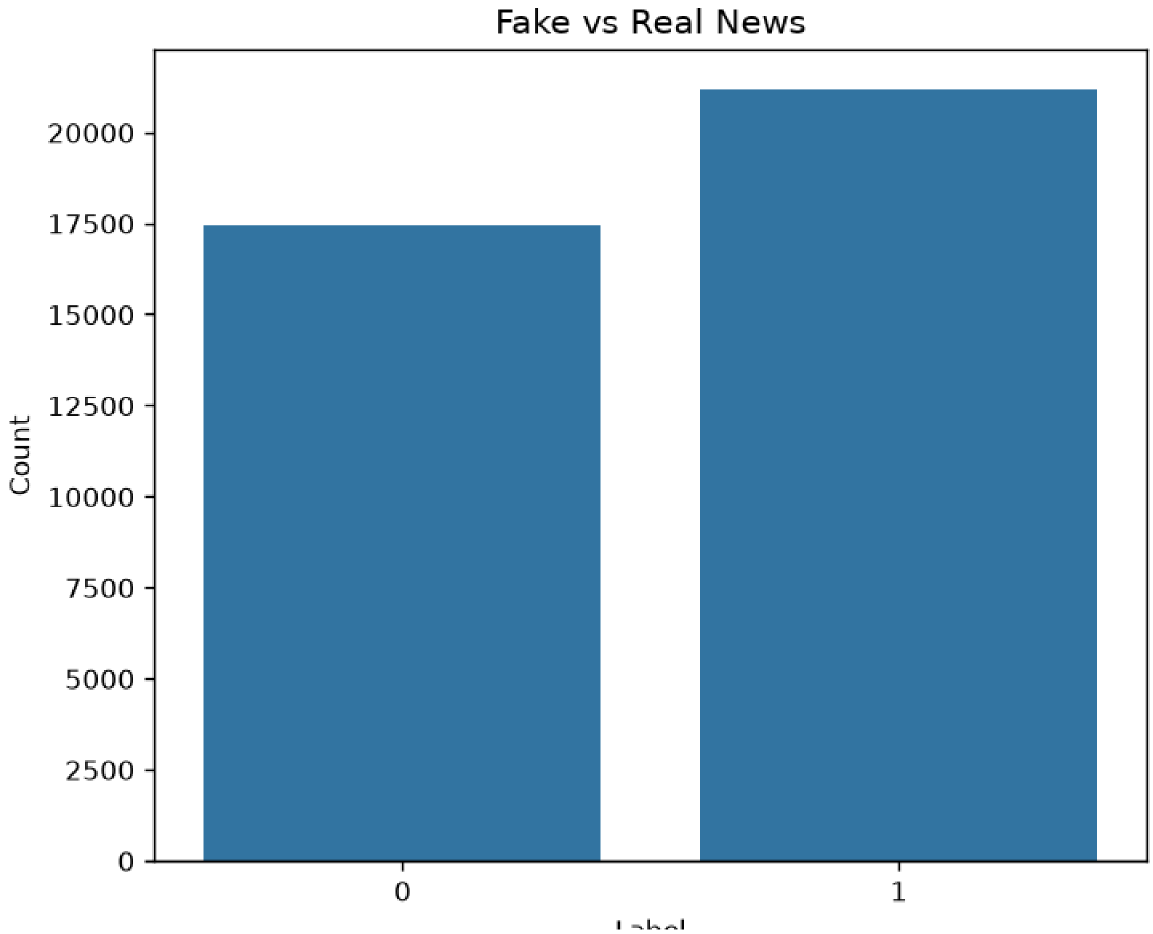
However, only the textual information was required for the classification task. Therefore, the **Title** and **Text** fields were concatenated to create a single feature representing the complete news article, while the **Subject** and **Date** columns were excluded because they were not directly involved in textual classification.

The final dataset therefore consisted of two primary attributes:

- Cleaned News Text
- Label (0 = Fake, 1 = Real)

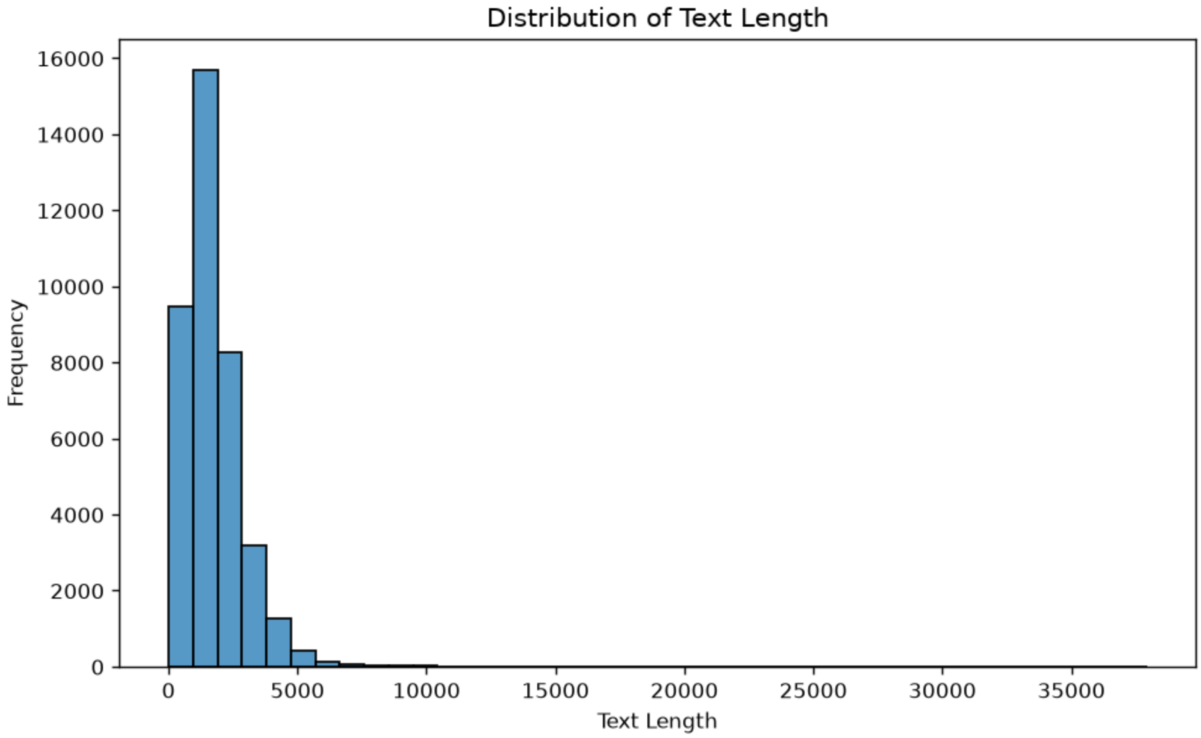
The dataset inspection performed during Phase 1 indicated that there were **no missing values** in any of the columns, eliminating the need for missing value imputation. Furthermore, the label distribution demonstrated that the dataset remained relatively balanced after preprocessing, containing **21,191 real news articles** and **17,455 fake news articles**, which helps reduce prediction bias toward a particular class.

Figure 1 illustrates the distribution of fake and real news articles after preprocessing. The relatively balanced distribution ensures that the trained machine learning models receive sufficient examples from both categories during learning.(Figure 1 as shown->)



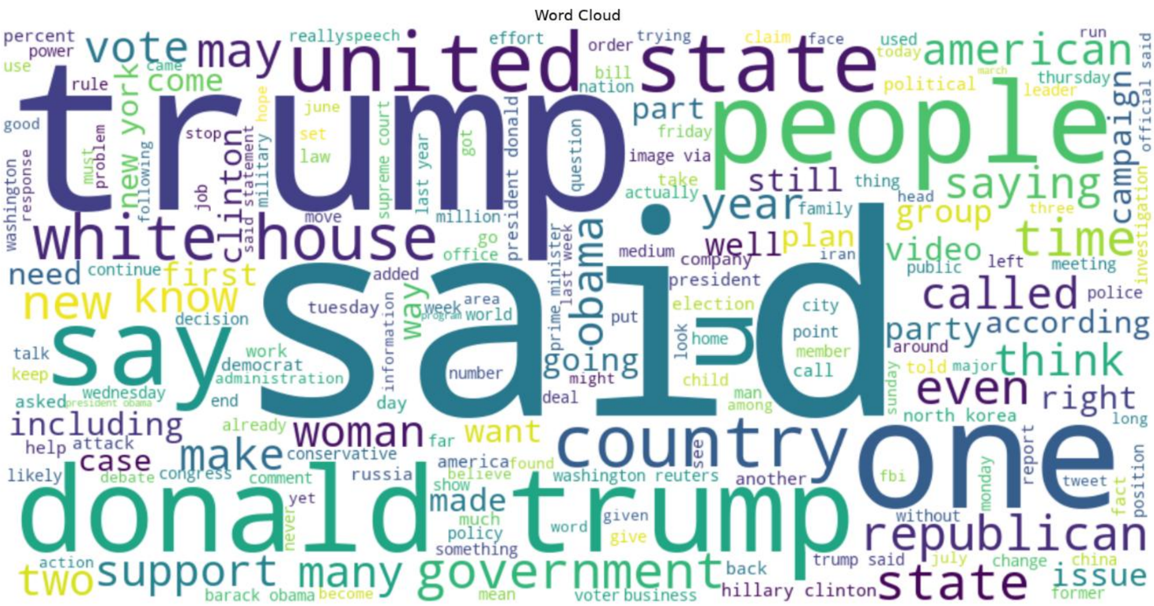
In addition to label analysis, exploratory data analysis was performed to understand the characteristics of the textual data. The distribution of text length revealed that most news articles contain between one thousand and three thousand processed characters, while only a small number of articles exceed this range.

(Figure 2: Distribution of Text Length)



Word frequency analysis was also performed to identify the most common words appearing in the corpus. The generated Word Cloud indicates that political terms such as **Trump**, **State**, **Donald**, **People**, **Republican**, **Government**, and **White House** occur frequently because the dataset mainly consists of political news published during the 2016–2017 period.

(Figure 3: Word Cloud)



The statistical summary generated during preprocessing confirmed that the dataset contains sufficient textual diversity for machine learning while maintaining high data quality without null values or duplicate news articles.

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3. Methodology

The proposed fake news detection system follows a complete Natural Language Processing and Machine Learning pipeline. The methodology consists of six major stages:

1. Dataset Preparation
2. Text Preprocessing
3. Feature Extraction
4. Model Training
5. Model Evaluation
6. Prediction System

The overall workflow begins with loading the dataset, followed by cleaning and preprocessing the textual content. The processed text is converted into numerical feature vectors using TF-IDF and Bag-of-Words representations before being supplied to multiple supervised machine learning classifiers. The trained models are then evaluated using various performance metrics, and the best-performing classifier is deployed for real-time news prediction.

3.1 Data Preprocessing

Raw textual data collected from online news sources usually contains irrelevant symbols, HTML tags, URLs, numbers, punctuation marks, and stop words that do not contribute significantly to classification performance. Therefore, preprocessing is an essential stage that improves both computational efficiency and predictive accuracy.

The preprocessing pipeline implemented in this project consists of the following steps:

Lowercase Conversion

All characters were converted into lowercase to ensure that words such as "Government" and "government" were treated as identical terms.

URL Removal

Hyperlinks appearing inside news articles were removed using regular expressions because URLs do not contribute meaningful semantic information for classification.

HTML Tag Removal

Any HTML formatting present within the dataset was eliminated before further processing.

Number Removal

Numerical values were removed because they generally contribute less toward distinguishing fake and genuine news articles.

Punctuation Removal

All punctuation symbols were removed to simplify tokenization and reduce vocabulary size.

Tokenization

Each cleaned sentence was divided into individual words using the Natural Language Toolkit (NLTK) tokenizer.

Stop Word Removal

Common English words such as *the*, *is*, *and*, *was*, and *of* were removed because they occur frequently but contribute little discriminative information.

Lemmatization

Finally, words were converted into their base grammatical forms using WordNet Lemmatizer. For example:

- Running → Run
- Studies → Study
- Governments → Government

Lemmatization reduces vocabulary size while preserving semantic meaning, thereby improving feature quality.

3.2 Exploratory Data Analysis (EDA)

Before model training, exploratory data analysis was performed to understand the characteristics of the dataset. Several visualizations were generated, including:

- Label Distribution
- Text Length Distribution
- Word Cloud
- Top Twenty Frequent Words

These analyses provide insights into the vocabulary and structural properties of the news corpus and help verify that preprocessing has been performed correctly.

3.3 Feature Extraction

Machine learning algorithms cannot directly process textual information. Therefore, the cleaned text must first be converted into numerical feature vectors.

Two feature extraction techniques were implemented:

TF-IDF

The primary feature extraction technique used throughout the project was **Term Frequency–Inverse Document Frequency (TF-IDF)** with a maximum vocabulary size of **10,000 features** and both unigram and bigram representations.

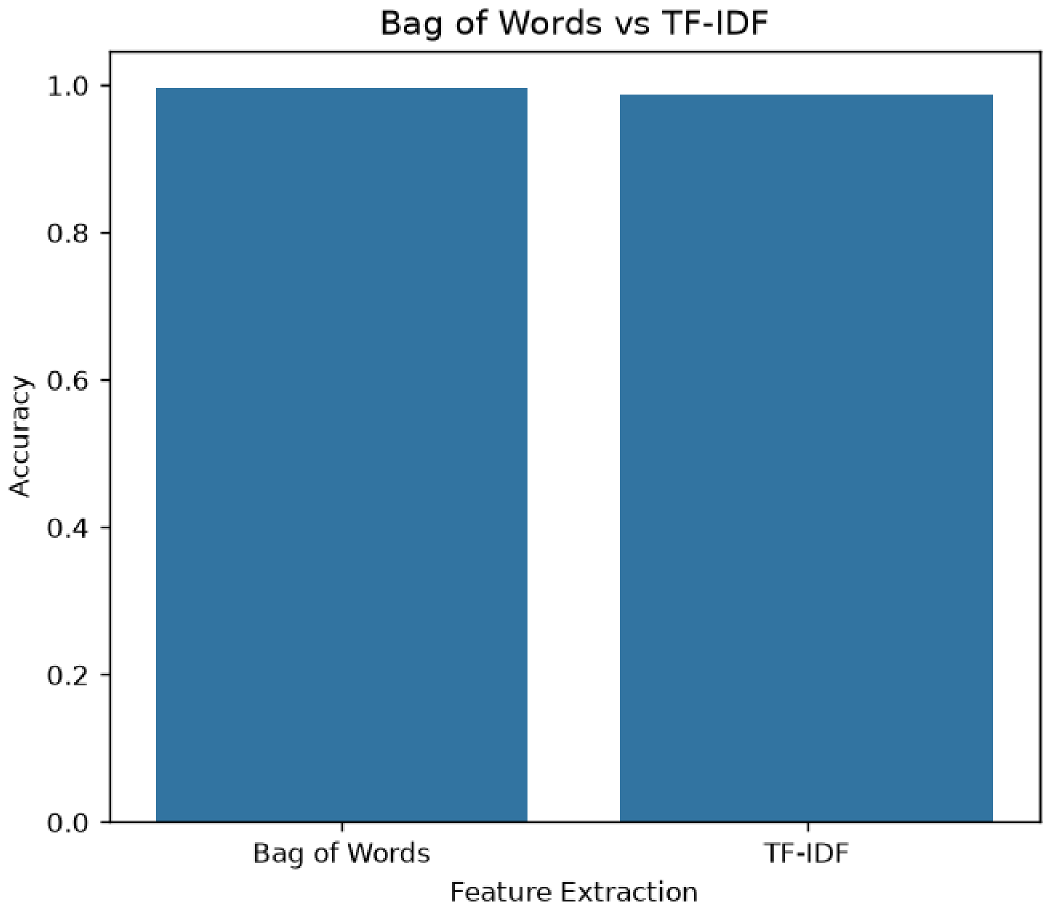
TF-IDF assigns higher weights to words that are important within a particular document while reducing the influence of frequently occurring words across the entire corpus. This enables the classifier to identify informative terms more effectively than simple frequency counts.

Bag of Words

To compare feature extraction methods, a Bag-of-Words model using **5,000 features** was also implemented. Bag-of-Words represents each document based solely on word occurrence frequencies without considering the relative importance of words.

The comparative analysis performed during Phase 3 demonstrated that the Bag-of-Words representation achieved an accuracy of approximately **99.59%**, while the TF-IDF representation achieved approximately **98.75%** using Logistic Regression.

(Figure 4: Bag of Words vs TF-IDF Accuracy Comparison)



4. Results

The primary objective of this project was to develop an efficient machine learning system capable of distinguishing between fake and genuine news articles using Natural Language Processing techniques. To evaluate the effectiveness of the proposed approach, four supervised machine learning algorithms were trained and tested using identical training and testing datasets. The models considered in this study include K-Nearest Neighbors (KNN), Logistic Regression, Random Forest, and Multi-Layer Perceptron (MLP) Neural Network.

The dataset was divided into training and testing subsets using an 80:20 ratio while maintaining class balance through stratified sampling. After preprocessing and TF-IDF feature extraction, each classifier was trained independently and evaluated using multiple performance metrics instead of relying solely on classification accuracy. The evaluation metrics included Accuracy, Precision, Recall, F1-score, Confusion Matrix, Receiver Operating Characteristic (ROC) Curve, Precision-Recall Curve, Cross Validation, and Feature Importance Analysis. This comprehensive evaluation provides a better understanding of each model's strengths and weaknesses.

The experimental results indicate that ensemble learning techniques significantly outperform distance-based classification methods for fake news detection. Among all evaluated algorithms, the Random Forest classifier achieved the highest prediction accuracy, followed closely by the Neural Network and Logistic Regression models.

Table 1 summarizes the performance of all four machine learning models.

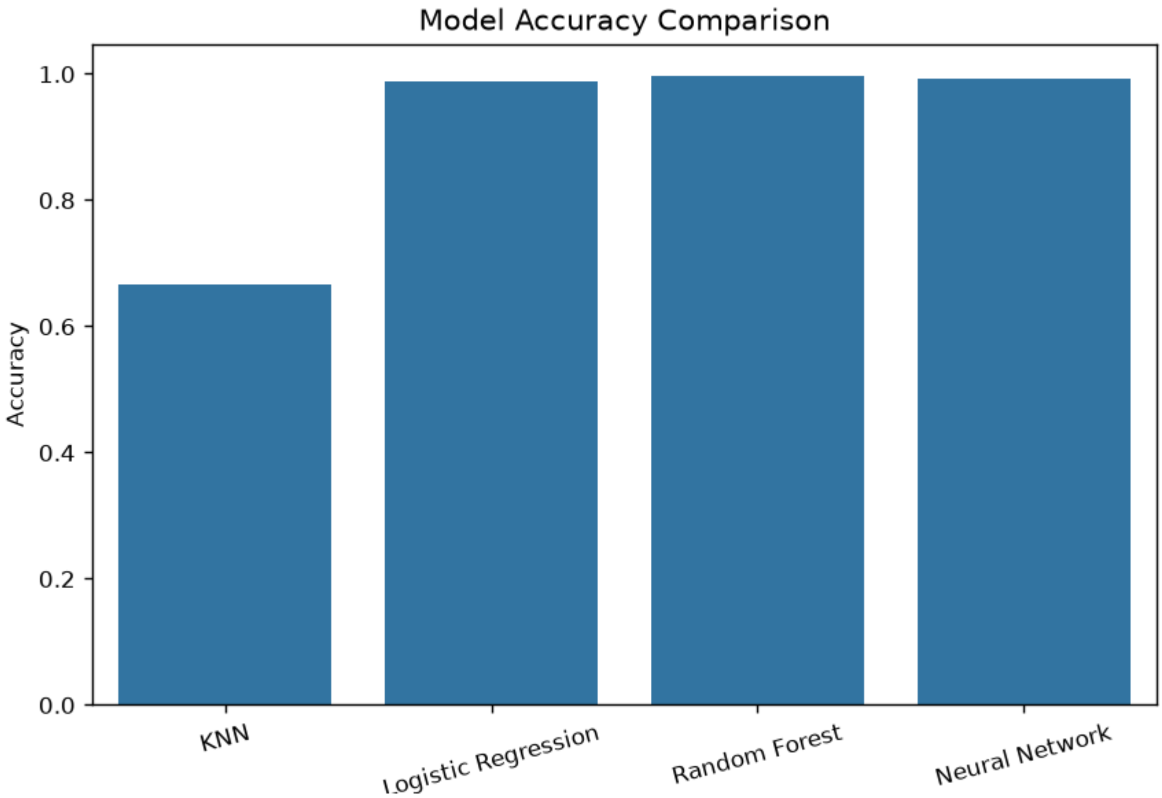
Table 1. Performance Comparison of Machine Learning Models

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
K-Nearest Neighbors	66.68%	95.51%	41.17%	57.53%
Logistic Regression	98.75%	98.43%	99.29%	98.86%
Random Forest	**99.66%**	**99.51%**	**99.88%**	**99.69%**
Neural Network	99.29%	99.11%	99.60%	99.35%

The comparison clearly demonstrates that Random Forest provides the highest overall classification performance. Logistic Regression also performs remarkably well despite being a comparatively simple linear classifier. In contrast, KNN exhibits significantly lower recall and overall accuracy because its predictions depend heavily on the nearest neighboring samples within a high-dimensional TF-IDF feature space.

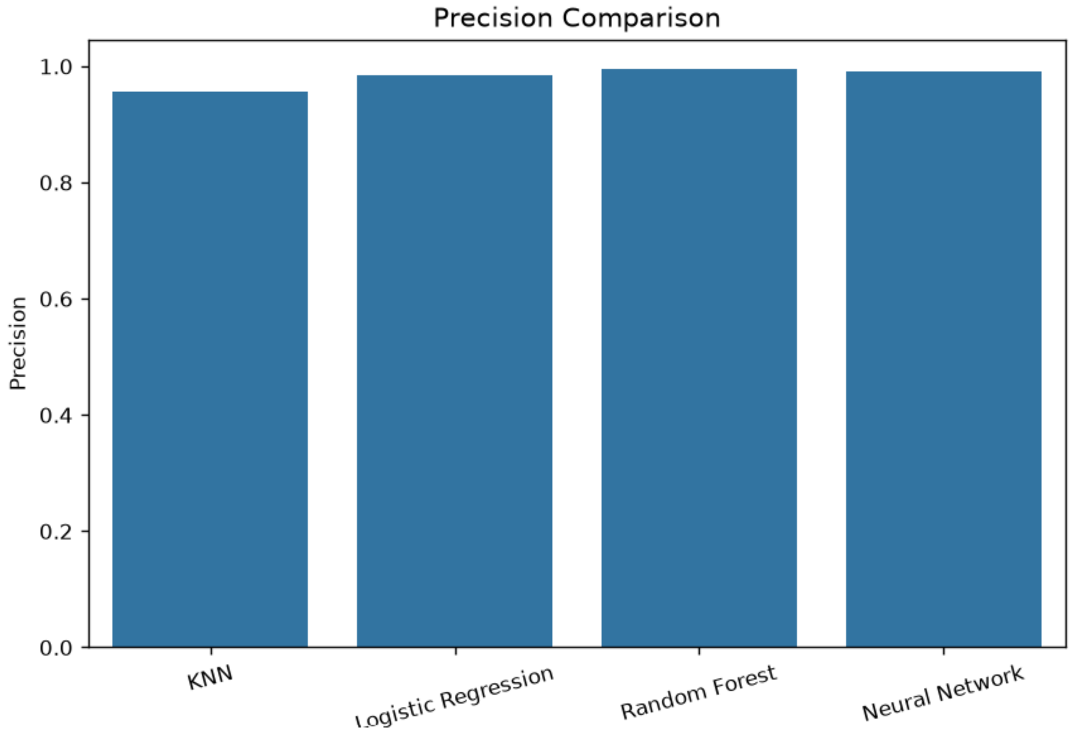
The comparison of model accuracies is illustrated in **Figure 5**, where Random Forest achieves the highest accuracy among all classifiers.

(Figure 5: Model Accuracy Comparison)

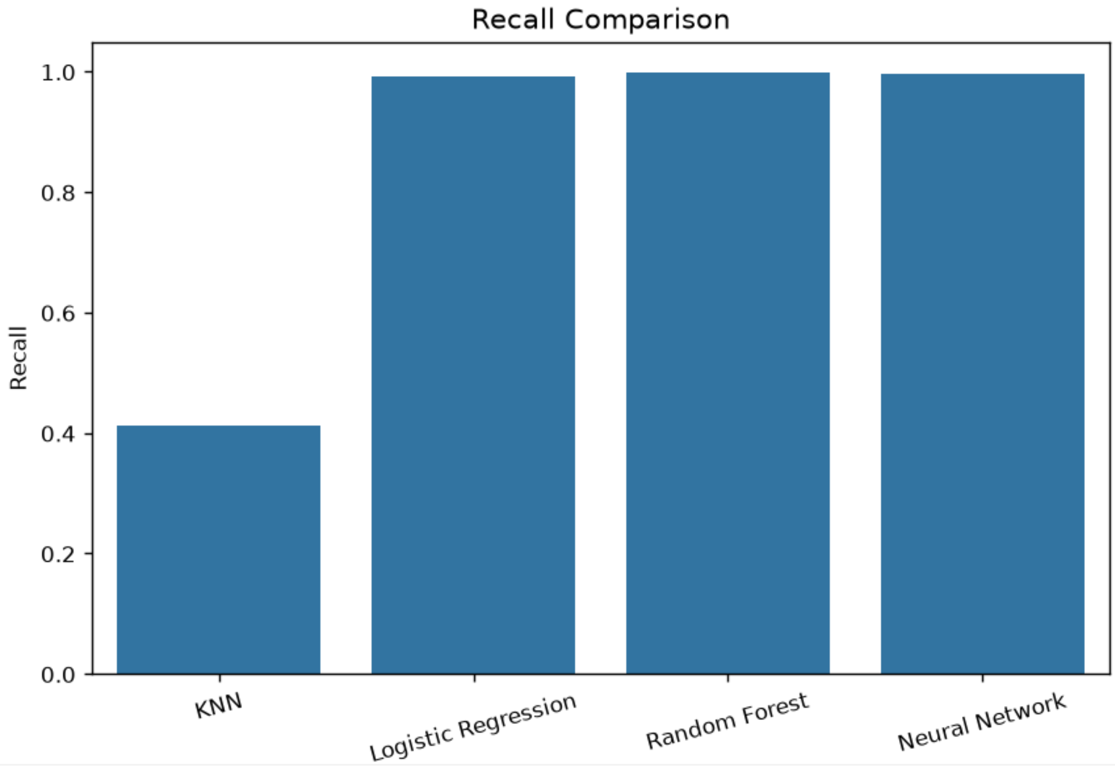


Similarly, **Figures 6, 7, and 8** compare Precision, Recall, and F1-score, respectively. The graphical comparison confirms that Random Forest consistently provides the highest values across all evaluation metrics, whereas KNN performs considerably worse due to poor recall.

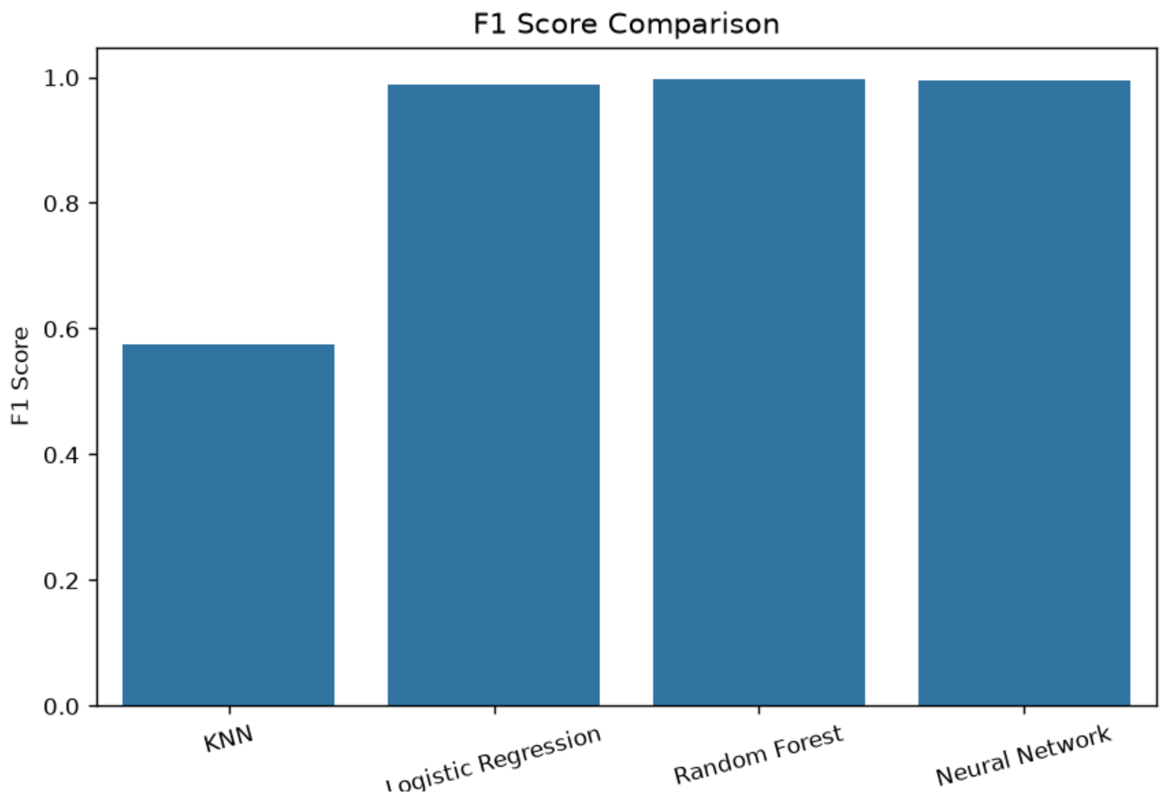
(Figure 6: Precision Comparison)



(Figure 7: Recall Comparison)



(Figure 8: F1-Score Comparison)



4.1 Confusion Matrix Analysis

A confusion matrix provides detailed information regarding correctly and incorrectly classified news articles. Unlike overall accuracy, the confusion matrix allows visualization of True Positives, True Negatives, False Positives, and False Negatives separately.

Logistic Regression

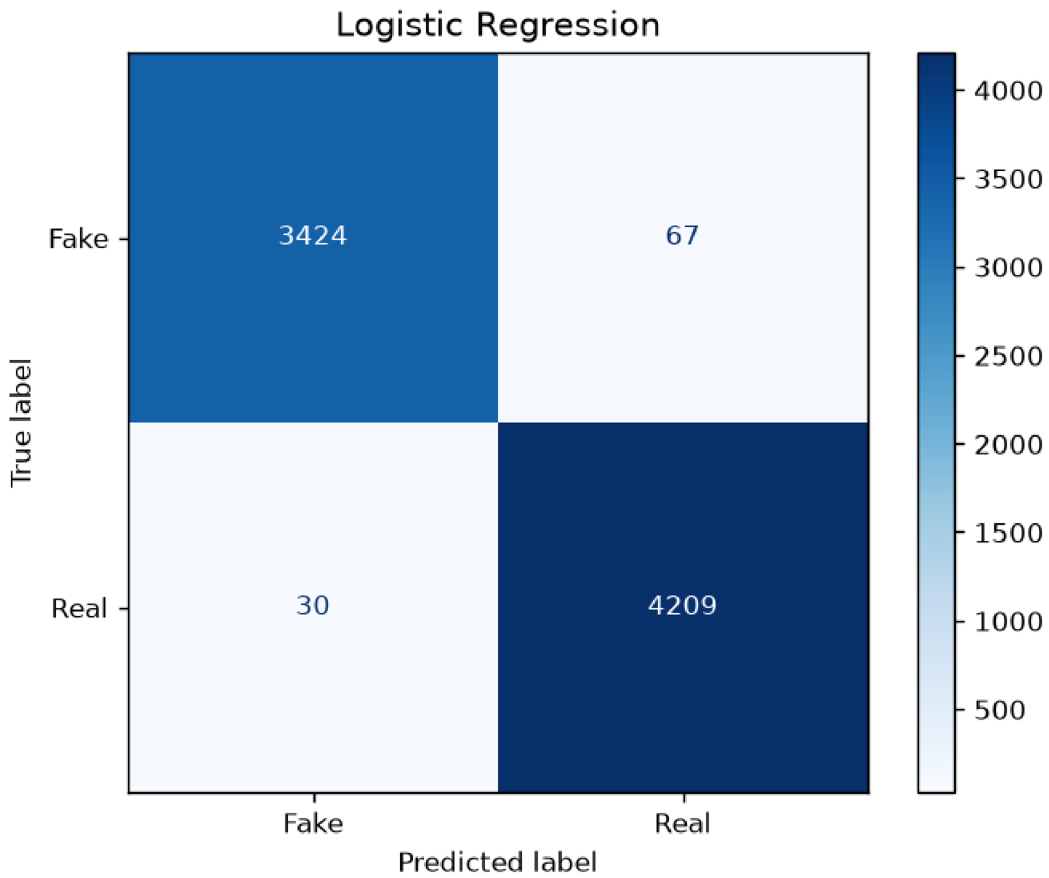
The Logistic Regression classifier correctly identified most fake and genuine news articles. Out of all testing samples, only a very small number of fake news articles were misclassified as genuine, while only a few genuine news articles were incorrectly classified as fake.

The confusion matrix shows:

- True Fake = **3424**
- False Real = **67**
- False Fake = **30**
- True Real = **4209**

The high diagonal values indicate excellent classification performance with very few prediction errors.

(Figure 9: Logistic Regression Confusion Matrix)



K-Nearest Neighbors

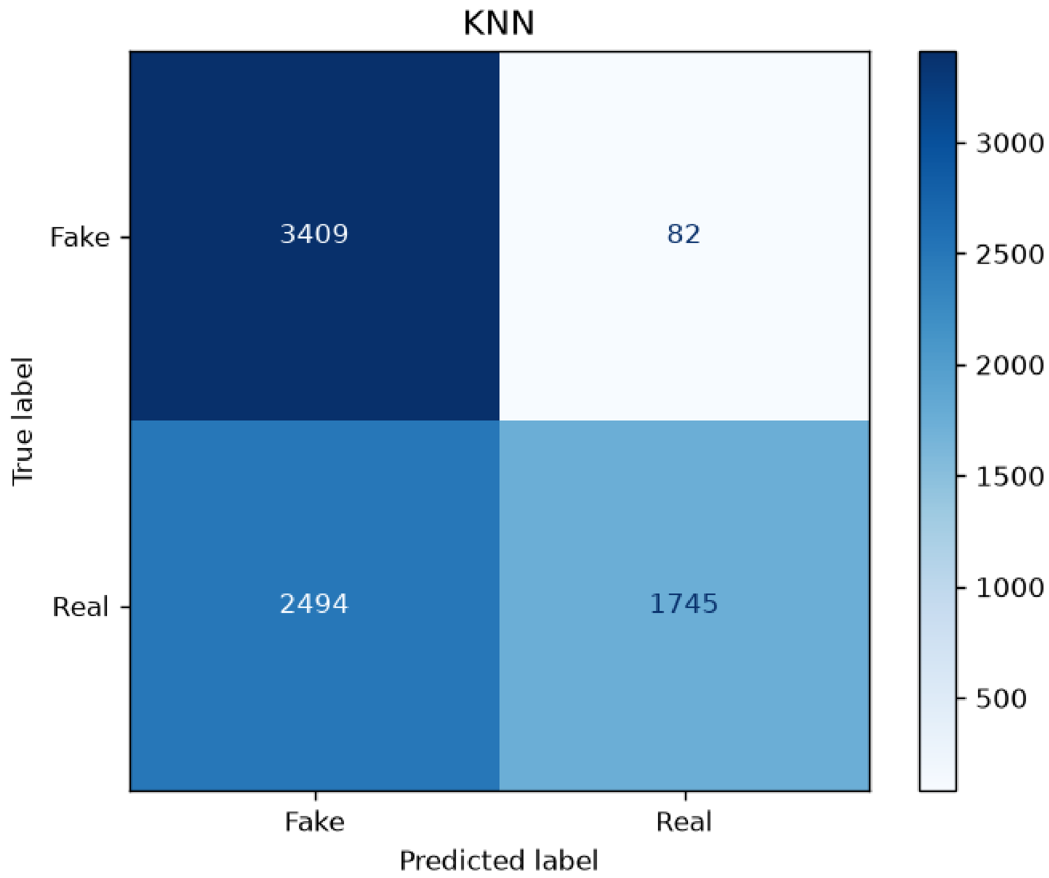
Among all evaluated models, KNN produced the weakest performance. Although it correctly identified many fake news articles, it failed to recognize a large number of genuine news articles.

The confusion matrix indicates:

- True Fake = **3409**
- False Real = **82**
- False Fake = **2494**
- True Real = **1745**

The large number of false negatives significantly reduces recall and overall classification accuracy.

(Figure 10: KNN Confusion Matrix)



Random Forest

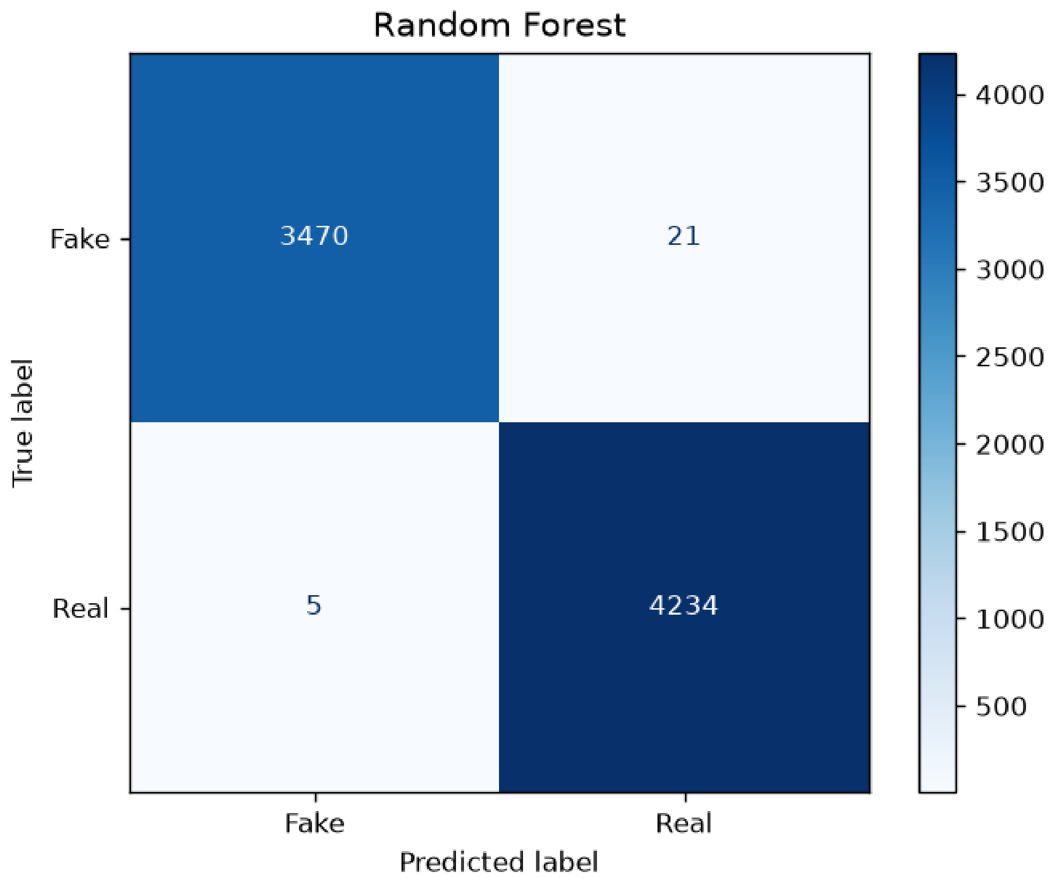
Random Forest produced the best overall classification performance. Almost every testing sample was classified correctly.

The confusion matrix demonstrates:

- True Fake = **3470**
- False Real = **21**
- False Fake = **5**
- True Real = **4234**

Only twenty-six prediction errors occurred among more than seven thousand testing samples, indicating exceptional model performance.

(Figure 11: Random Forest Confusion Matrix)



Neural Network

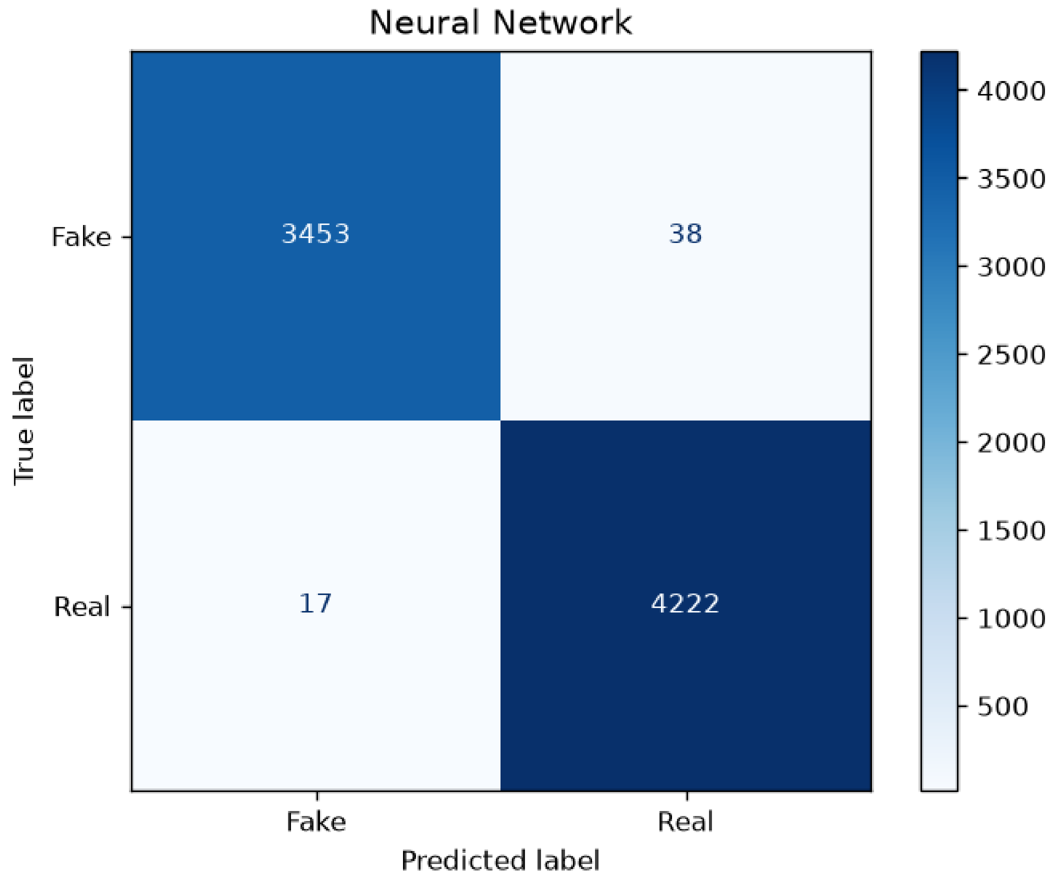
The Multi-Layer Perceptron Neural Network also achieved outstanding classification performance.

The confusion matrix contains:

- True Fake = **3453**
- False Real = **38**
- False Fake = **17**
- True Real = **4222**

Although its performance is slightly below Random Forest, it still demonstrates excellent classification capability.

(Figure 12: Neural Network Confusion Matrix)



4.2 ROC Curve Analysis

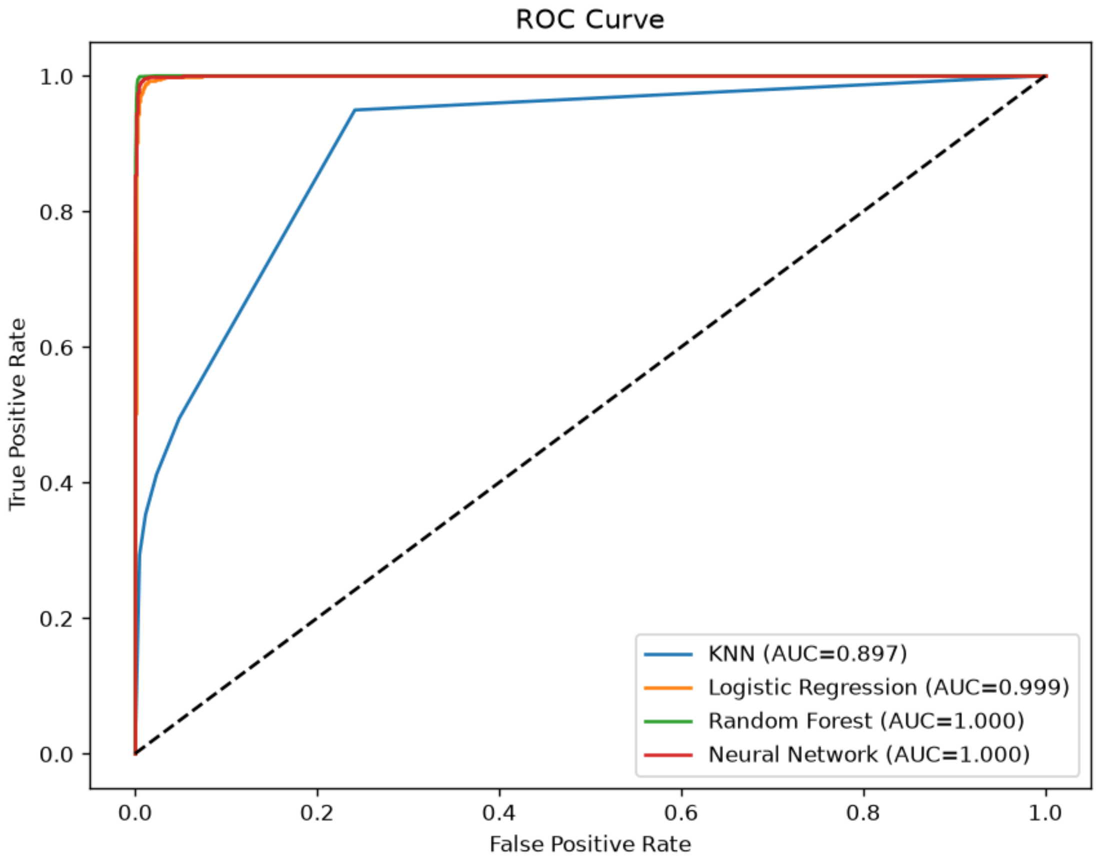
Receiver Operating Characteristic (ROC) analysis evaluates the trade-off between the True Positive Rate and False Positive Rate. The Area Under the Curve (AUC) provides an overall measure of classifier performance.

The obtained AUC values are:

Model	AUC
KNN	0.897
Logistic Regression	0.999
Random Forest	1.000
Neural Network	1.000

The ROC curve illustrates that Random Forest and Neural Network almost perfectly separate fake and genuine news articles. Logistic Regression also achieves an almost ideal ROC curve, while KNN shows comparatively weaker performance.

(Figure 13: ROC Curve)

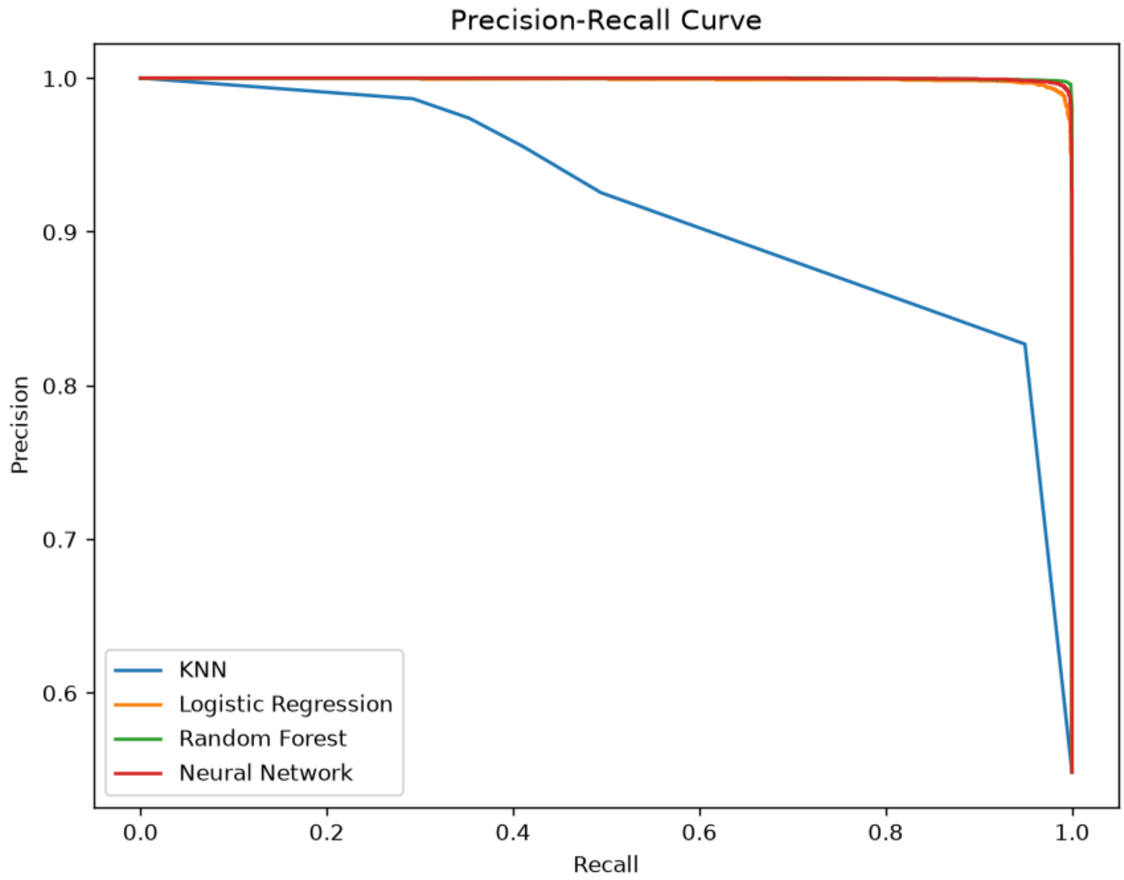


4.3 Precision–Recall Curve

Precision–Recall analysis is particularly useful for binary classification problems involving misinformation detection because it evaluates how accurately the classifier identifies positive samples while minimizing false predictions.

The Precision–Recall curves demonstrate that Logistic Regression, Random Forest, and Neural Network maintain precision values close to one throughout nearly the entire recall range. KNN, however, experiences a noticeable decline in precision as recall increases, further confirming its comparatively weaker classification capability.

(Figure 14: Precision–Recall Curve)



4.4 Feature Importance Analysis

To understand which textual features contribute most significantly to classification, feature importance analysis was performed using the trained Random Forest classifier.

The twenty most influential words extracted by the model include:

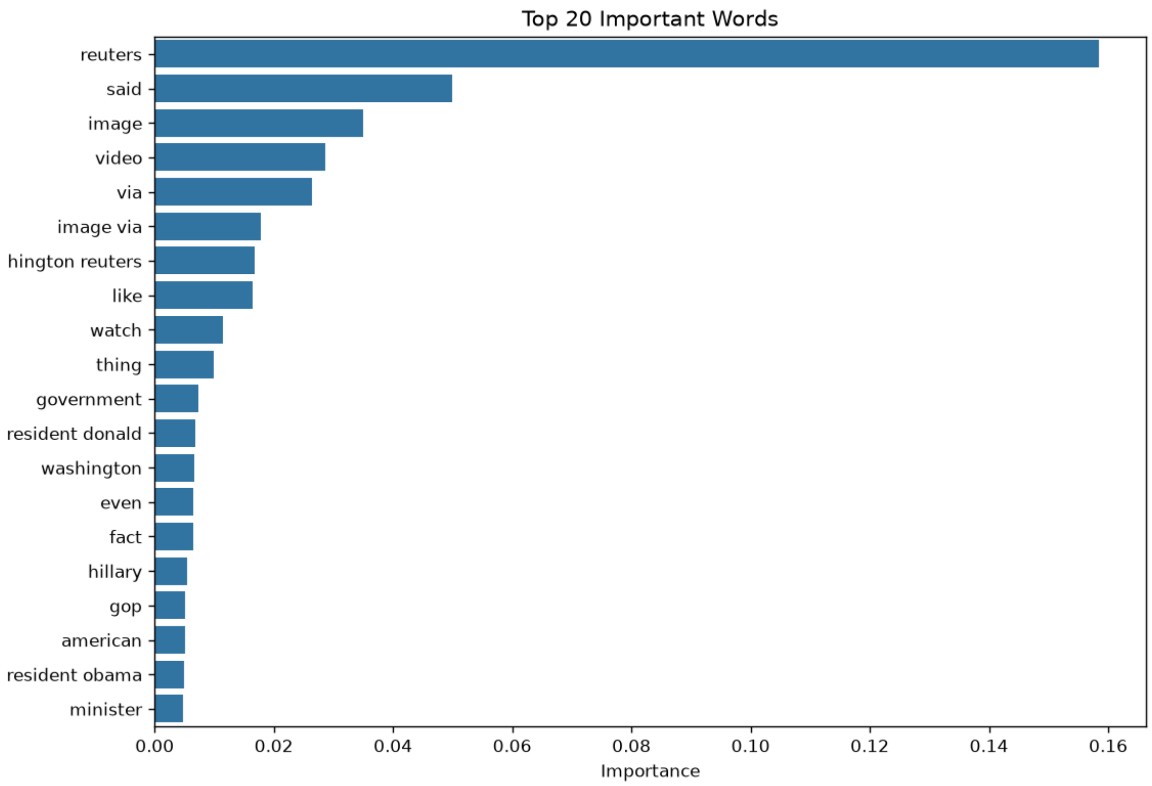
- Reuters
- Said
- Image
- Video
- Via
- Government
- Washington
- Donald
- Hillary

- Obama
- GOP
- Minister

Among all extracted features, **"Reuters"** possesses the highest importance score. This observation is expected because the genuine news articles contained within the dataset primarily originate from Reuters. Consequently, the classifier learns to associate this term strongly with genuine news.

Other politically significant terms such as **Government**, **Washington**, **Donald**, and **Obama** also contribute substantially toward classification because the dataset consists largely of political news published during the 2016–2017 period.

(Figure 15: Top 20 Important Words)



4.5 Feature Extraction Comparison

To evaluate the impact of feature extraction techniques, both Bag-of-Words and TF-IDF representations were implemented using Logistic Regression.

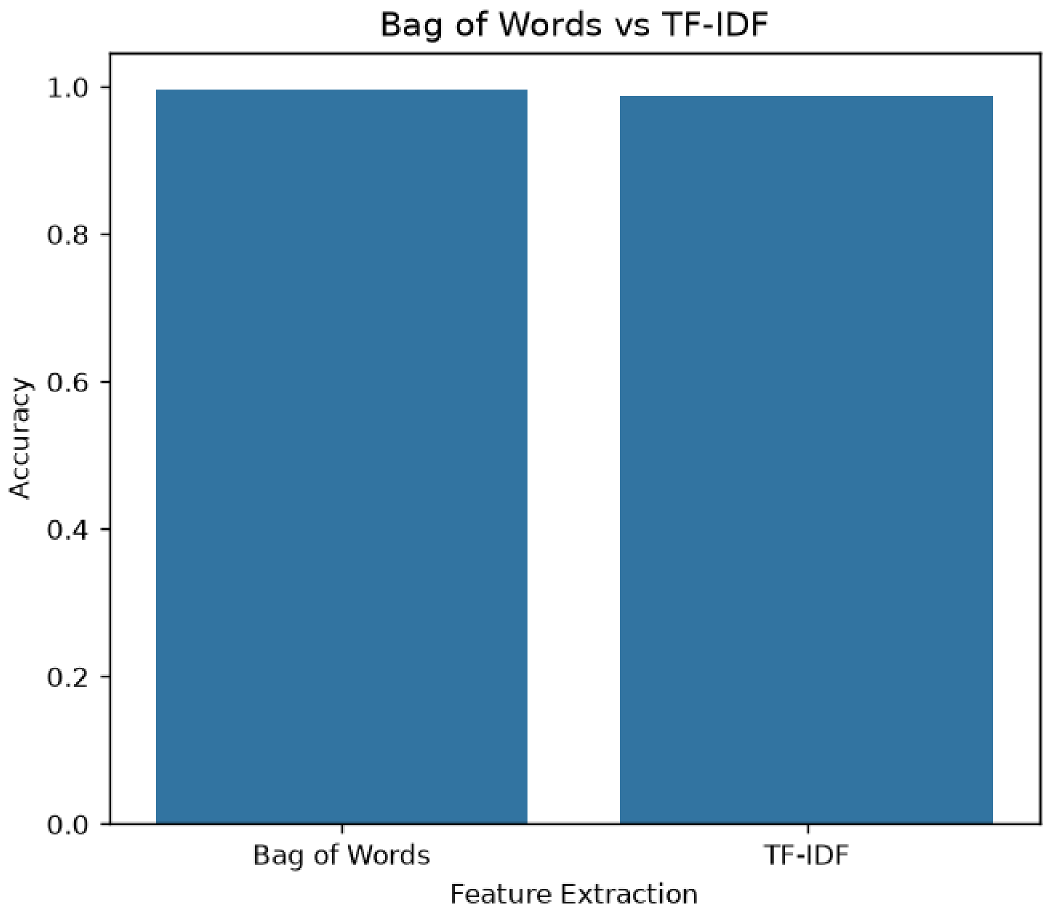
The experimental comparison shows:

- Bag-of-Words Accuracy = **99.59%**

- TF-IDF Accuracy = **98.75%**

Although both methods produced excellent results, Bag-of-Words achieved a slightly higher classification accuracy for this dataset.

(Figure 16: Bag of Words vs TF-IDF Accuracy Comparison)



4.6 Cross-Validation Results

To evaluate the stability and generalization capability of the trained models, three-fold cross-validation was performed.

The average accuracies obtained are:

Model	**Cross-Validation Accuracy**
KNN	64.38%
Logistic Regression	98.70%
Random Forest	**99.57%**

Model	Cross-Validation Accuracy
Neural Network	99.12%

These results indicate that the Random Forest classifier consistently achieves the highest accuracy across different data partitions, demonstrating strong generalization performance and minimal overfitting.

5. Discussion

The primary objective of this project was to compare the performance of different supervised machine learning algorithms for fake news detection using Natural Language Processing techniques. Four classification algorithms—K-Nearest Neighbors (KNN), Logistic Regression, Random Forest, and Multi-Layer Perceptron Neural Network—were implemented and evaluated under identical preprocessing and feature extraction conditions. The experimental results clearly demonstrate that the choice of classifier has a significant impact on the effectiveness of fake news detection.

Among all evaluated models, the Random Forest classifier achieved the highest performance across almost every evaluation metric. It produced an accuracy of approximately **99.66%**, precision of **99.51%**, recall of **99.88%**, and F1-score of **99.69%**, making it the most reliable classifier for this dataset. The confusion matrix further confirmed that only a very small number of news articles were incorrectly classified, indicating excellent generalization capability.

The Logistic Regression classifier also produced excellent results, achieving an accuracy of approximately **98.75%**. Logistic Regression is considered a **parametric machine learning model** because it assumes a predefined mathematical relationship between the input features and the output class. Despite its relatively simple mathematical structure, Logistic Regression performed remarkably well when combined with TF-IDF feature extraction. Its computational efficiency, simplicity, and interpretability make it a strong baseline for text classification problems.

The Multi-Layer Perceptron Neural Network achieved an accuracy of approximately **99.29%**, demonstrating its ability to learn complex nonlinear relationships within textual data. Neural networks generally require more computational resources than traditional classifiers, but they are capable of capturing sophisticated feature interactions that simpler algorithms may overlook. In this project, the Neural Network performed competitively with Random Forest and Logistic Regression, although its performance remained slightly below that of the Random Forest classifier.

The K-Nearest Neighbors classifier exhibited significantly weaker performance compared with the other three models. Although its precision remained relatively high, its recall value was considerably lower, resulting in an overall accuracy of approximately **66.68%**. KNN is a **non-parametric algorithm**, meaning it does not assume any predefined mathematical model and instead makes predictions based on similarity between neighboring samples. While this approach can perform well on low-dimensional datasets, it becomes less effective in high-dimensional feature spaces such as TF-IDF representations, where distance calculations become less meaningful. Consequently, KNN struggled to correctly classify many genuine news articles, leading to a large number of false negatives.

The comparative analysis between parametric and non-parametric models provides several important observations.

Parametric Models

Parametric models used in this project include Logistic Regression and the Neural Network. These models learn mathematical relationships during training and then apply those learned parameters to classify unseen data. Their advantages include:

- Faster prediction speed.
- Better scalability for large datasets.
- Efficient handling of high-dimensional TF-IDF features.
- Consistent classification performance.

However, parametric models may require careful parameter tuning and sufficient training data to achieve optimal performance.

Non-Parametric Models

The primary non-parametric model evaluated in this project is K-Nearest Neighbors. Instead of learning explicit mathematical parameters, KNN stores the training data and compares new observations directly with previously seen samples.

Advantages of non-parametric models include:

- Simple implementation.
- No assumptions regarding underlying data distribution.
- Ability to model complex decision boundaries.

However, the disadvantages observed during this project include:

- Increased computational cost during prediction.
- Poor scalability for large datasets.
- Reduced effectiveness in high-dimensional feature spaces.
- Lower recall and overall classification accuracy.

Random Forest represents an ensemble learning approach that combines multiple decision trees. Although decision trees themselves are generally considered non-parametric models, the ensemble structure enables Random Forest to reduce overfitting while improving predictive performance. The experimental results indicate that Random Forest successfully balances bias and variance, allowing it to outperform both KNN and Logistic Regression.

Another important observation concerns the influence of the dataset itself. The generated Word Cloud and Feature Importance Analysis reveal that the dataset is dominated by political news articles containing terms such as "Reuters," "Trump," "Government," "Washington," and "Republican." As a result, the trained models become particularly effective at classifying political news. However, prediction experiments performed on technology, healthcare, and scientific news articles demonstrated that some genuine articles not closely resembling the training distribution may receive lower confidence scores. This behaviour highlights one limitation of traditional machine learning approaches based on TF-IDF features: they rely heavily on vocabulary learned during training and may struggle to generalize to entirely new writing styles or subject domains.

Overall, the experimental evaluation confirms that combining effective text preprocessing with TF-IDF feature extraction and Random Forest classification provides a highly accurate solution for fake news detection.

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6. Conclusion

This project successfully developed an AI-powered fake news detection system using Natural Language Processing and supervised machine learning techniques. The system performs the complete workflow required for automated fake news detection, including data loading, preprocessing, exploratory data analysis, feature extraction, machine learning model training, performance evaluation, and prediction of unseen news articles.

The preprocessing stage significantly improved data quality by removing unnecessary textual information such as URLs, HTML tags, punctuation, numbers, and stop words while applying tokenization and lemmatization to normalize the textual content. Feature extraction using TF-IDF converted the processed news articles into meaningful numerical representations suitable for machine learning algorithms.

Four supervised learning algorithms were implemented and compared. Among these algorithms, Random Forest achieved the highest overall performance, followed closely by the Neural Network and Logistic Regression models. K-Nearest Neighbors produced comparatively weaker results because of the high-dimensional nature of TF-IDF feature vectors.

The experimental evaluation demonstrated that Random Forest achieved approximately **99.66% classification accuracy**, with excellent precision, recall, and F1-score values. Cross-validation, ROC analysis, Precision-Recall analysis, and Feature Importance Analysis further confirmed the robustness and reliability of the proposed system.

The developed prediction module successfully classifies user-provided news articles into **REAL NEWS** or **FAKE NEWS** while also displaying the associated prediction probabilities. This functionality demonstrates the practical applicability of the proposed system for automated misinformation detection.

Although the proposed system performs exceptionally well on the selected dataset, several limitations remain. The dataset primarily consists of political news articles collected during a limited time period, reducing the model's ability to generalize to technology, healthcare, sports, entertainment, or multilingual news. Furthermore, TF-IDF-based feature extraction cannot capture deeper semantic relationships between words in the same manner as modern transformer-based language models.

Future work may involve integrating contextual language models such as **BERT**, **RoBERTa**, or **DistilBERT**, incorporating multilingual datasets, developing explainable AI techniques for prediction interpretation, and deploying the trained model as a web application or browser extension capable of performing real-time fake news verification.

Overall, the project demonstrates that machine learning and Natural Language Processing provide an effective, scalable, and highly accurate solution for combating the growing challenge of online misinformation.

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7. Appendix

The appendix contains the implementation details and supporting materials developed during this project.

Appendix A – Source Code

The complete Python implementation consists of three major phases:

Phase 1

- Dataset Loading
- Data Cleaning
- Text Preprocessing
- Exploratory Data Analysis

- Word Cloud Generation
- Label Distribution
- Text Length Distribution

Phase 2

- TF-IDF Feature Extraction
- Train-Test Split
- K-Nearest Neighbors
- Logistic Regression
- Random Forest
- Multi-Layer Perceptron Neural Network
- Confusion Matrices
- Accuracy Comparison
- Precision Comparison
- Recall Comparison
- F1-score Comparison

Phase 3

- Bag-of-Words Feature Extraction
- Feature Importance Analysis
- Model Saving using Joblib
- Prediction Module
- ROC Curve
- Precision-Recall Curve
- Cross Validation
- GridSearchCV Hyperparameter Optimization
- Exporting Results

Appendix B – Generated Files

The following files are generated in the project:

- cleaned_fake_news.csv
- model_comparison.csv
- feature_importance.csv
- fake_news_model.pkl
- tfidf_vectorizer.pkl

Appendix C – Experimental Outputs

The following figures generated during execution:

1. Fake vs Real News Distribution
2. Text Length Distribution
3. Word Cloud
4. Logistic Regression Confusion Matrix
5. KNN Confusion Matrix
6. Random Forest Confusion Matrix
7. Neural Network Confusion Matrix
8. Accuracy Comparison
9. Precision Comparison
10. Recall Comparison
11. F1-score Comparison
12. ROC Curve
13. Precision-Recall Curve
14. Top 20 Important Words
15. Bag-of-Words vs TF-IDF Comparison

References (IEEE Format)

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